

POSSIBLE PROTECTIVE ANTICANCER EFFECT OF ETHYL ACETATE FRACTION OF IRAQI *HIBISCUS TILIACEUS* L. LEAVES EXTRACT ON DIETHYLNITROSAMINE-INDUCED HEPATOCARCINOGENESIS IN MALE RATS

ZAHRAA MAAN ABDUL-AZEEZ ^{1*}, SHIHAB HATTAB MUTLAG ²

¹Medicolegal Directorate, Ministry of Health and Environment, Baghdad-Iraq.

²Department of Pharmacology and Toxicology, College of Pharmacy, University of Baghdad, Baghdad, Iraq.

*corresponding author: zahraa89atos@gmail.com

Manuscript received: October 2023

Abstract

Hepatocellular carcinoma is a serious disease with high malignancy, rapid development and a poor prognosis. It starts with fibrosis, cirrhosis, and then cancer. Diethylnitrosamine (DEN) is a strong carcinogen used to induce cancer in animals. Due to their bioactive ingredients, herbal plants have been widely employed as low-cost, low-side-effect alternative medicines for liver problems. *Hibiscus tiliaceus* L. (Malvaceae) plant extracts contain flavonoids, coumarins, phenols, glycosides, tannins, triterpenoids, saponins, quinones and steroids. Thus, it offers several health benefits and medical and therapeutic qualities, including hepatoprotective, antimutagenic and natural antibacterial and antioxidant capabilities.

The study examined whether the ethyl acetate leaf extract of Iraqi *Hibiscus tiliaceus* L. in two graded dosages protected male Wistar Albino rats against diethylnitrosamine-induced liver cancer. In this study, rats were randomized into four groups (six rats per group): Group I (control group): received 1 mL/kg/day of distilled water for 20 weeks; Group II (HCC induction): received 70 mg/kg diethyl nitrosamine intraperitoneally once per week for ten weeks; Group III: received 250 mg/kg of the ethyl acetate leaf extract of *Hibiscus tiliaceus* orally with their food for five days per week for 20 weeks; and Group IV: rats received 500 mg/kg of the ethyl acetate leaf extract of *Hibiscus tiliaceus* as oral administration with their food for five days per week for 20 weeks. Iraqi *Hibiscus tiliaceus* L. ethyl acetate leaf extract significantly reduced MDA, AFP, AFP-L3, ALT, AST, TBS and inflammatory mediators (TNF- α , IL-6 and IL-1 β) ($p < 0.05$) and improved the antioxidant levels (GSH and SOD) significantly, making it chemoprotective.

Rezumat

Carcinomul hepatocelular este prezintă un nivel ridicat de malignitate, o dezvoltare rapidă și cu un prognostic slab, debutând cu fibroză, ciroză și apoi cancer. Dietilnitrozamina (DEN) este un carcinogen puternic utilizat pentru a induce cancerul la animale. Extractele din *Hibiscus tiliaceus* L. (Malvaceae) conțin flavonoide, cumarine, fenoli, glicozide, taninuri, triterpenoide, saponine, chinone și steroizi. Studiul a analizat efectul extractului acetat de etil din *Hibiscus tiliaceus* L. asupra cancerului hepatic indus de dietilnitrozamină. În acest studiu, șobolanii au fost repartizați aleatoriu în patru grupuri (șase șobolani pe grup): Grupul I (grupul de control): au primit 1 ml/kg/zi de apă distilată timp de 20 de săptămâni; Grupul II (inducerea HCC): au primit 70 mg/kg de dietilnitrozamină intraperitoneal o dată pe săptămână timp de zece săptămâni; Grupul III: au primit 250 mg/kg de extract de frunze de acetat de etil de *Hibiscus tiliaceus* pe cale orală cu hrana lor timp de cinci zile pe săptămână timp de 20 de săptămâni; și Grupul IV: șobolanii au primit 500 mg/kg de extract *Hibiscus tiliaceus* în acetat de etil, administrat pe cale orală, timp de cinci zile pe săptămână pentru 20 de săptămâni. Extractul din frunzele de *Hibiscus tiliaceus* L. în acetat de etil a redus semnificativ MDA, AFP, AFP-L3, ALT, AST, TBS și mediatorii inflamatori (TNF- α , IL-6 și IL-1 β) ($p < 0,05$) și a îmbunătățit semnificativ nivelurile de antioxidanți (GSH și SOD).

Keywords: Hepatocellular carcinoma, Diethylnitrosamine, *Hibiscus tiliaceus* L. (Malvaceae), Alpha-fetoprotein, Alpha-fetoprotein-L3

Introduction

The liver is responsible for numerous vital processes, including removing bile, glucose, fat, urea and endogenous and external waste from circulation [1]. Hepatocytes are metabolically active cells that increase oxidative stress when signaling pathways that promote cancer cell growth and survival are engaged [2, 3]. In its early stages, liver cancer is commonly asymptomatic and is usually

found when it is advanced, making a successful operation more difficult or impossible. Hepatocellular carcinoma (HCC), a malignant tumour, often known as liver cancer, consists of cells resembling hepatocytes but differing in their differentiation [4]. Numerous risk factors, including alcohol, toxic drugs, parasites, viruses and chemicals, induce chronic hepatitis and inflammation, which can speed up the development

of cirrhosis, fibrosis and HCC in their various stages [5]. Diethylnitrosamine (a nitroso family compound) is a chemical that can be found in water, tobacco smoke, agricultural chemicals, fried foods, cured meats, cheddar cheese, pharmaceutical and cosmetic products, and in trace amounts in baby bottle nipples. It is a well-known potent hepatocarcinogenic agent that promotes cancer in experimental animals. According to studies, DEN-induced hepatocellular cancer in rats resembles hepatocellular carcinoma in humans. Neoplasia advances in three stages: initiation, promotion and progression. This process is known as carcinogenesis. DEN, a repeatable carcinogen, causes lesions in rats that resemble many benign and malignant tumour forms in people [6]. The promutagenic adducts O-6-ethyl deoxyguanosine and O-4 and O-6-ethyl deoxythymidine are delivered by DEN. They cause DNA chain damage, depurination, or attachment to DNA and frequently result in DNA miscoding, which triggers the onset of liver carcinogenesis [7]. The disadvantages of traditional cancer chemotherapies and the benefits of more natural therapeutic approaches are the main drivers of the increased interest in complementary and alternative medicines [8]. The ability of numerous medicinal plants to treat cancer and reduce inflammation is currently being thoroughly studied. Herbal plants are commonly employed as affordable, efficient substitutes for pharmaceuticals with fewer side effects since they are a great source of secondary metabolites with anticancer potential [9]. *Hibiscus tiliaceus* L. (*Sea hibiscus*) is a species of flowering tree in the *Malvaceae* family. *H. tiliaceus* is distinguished by its bright yellow flowers with a deep crimson center that turns orange and finally red over the day before falling. The leaves are heart-shaped, and the tree's limbs have a tendency to curl with time. It has become extensively naturalised all over the tropical and subtropical parts of the planet. This species is consumed as a beverage (tea) and used as a decorative tree [10]. According to preliminary investigations, *H. tiliaceus* leaf extract has been utilised for medical purposes due to its anti-inflammatory, anti-tumour, anti-nociceptive and antioxidant capabilities. It also included phenols, flavonoids, tannins, glycosides and terpenoids [11]. Therefore, the goal of this study is to determine whether the ethyl acetate fraction of Iraqi *Hibiscus tiliaceus* L. leaf extract can prevent diethylnitrosamine-induced hepatocarcinogenesis in rats by examining how it affects the liver biochemical marker alpha-fetoprotein (AFP), the HCC marker, as well as oxidative stress, inflammatory mediators and histopathological liver architecture changes.

Materials and Methods

Chemicals

Diethyl nitrosamine (DEN) was purchased from Sigma Aldrich in St. Louis, USA, and linear chemicals SLU Spanish provided the standard rat test kits for alkaline phosphatase (ALP), total bilirubin (TBIL), aspartate aminotransferase (AST) and alanine aminotransferase (ALT). The following Elisa kits were bought from MyBioSource USA: rat TNF, interleukin-1 beta, interleukin-6, malondialdehyde, glutathione, superoxide dismutase, transforming growth factor beta, alpha-fetoprotein and alpha-fetoprotein-L3.

Plant collection and preparation of Leaf extract

The fresh leaves of *Hibiscus tiliaceus* L. (*Malvaceae*) were collected from the Master of Regents City, Karbala-Baghdad Rd., Karbala, in April 2022.

The leaves were dried in the shade at room temperature, ground into a fine powder using an electrical mill and weighed. The ground plant material (roughly 150 g) was then defatted separately with sufficient hexane solvent for 24 hours to remove chlorophyll and the hexane-soluble compounds like waxy material, after which it was placed in the thimble chamber of the Soxhlet apparatus and put through the Soxhlet extractor using solvent ethanol (80% v/v). The extract was filtered and concentrated at 30 - 40 °C in a rotary evaporator. Subsequently, the filtrate was partitioned using ethyl acetate to obtain a semi-solid material. In order to properly prepare the fraction for animal administration [12, 13].

Phytochemical Screening Tests

The extract is ready for qualitative phytochemical analyses using established techniques to detect the following components [11]:

Test for flavonoids: adding 5 mL of diluted ammonia solution and concentrated H₂SO₄ to the extract. The development of yellow colouring showed the presence of flavonoids.

Test for tannins: 2 mL of the extract was applied to one or two drops of 10% ferric chloride and a bluish-green colour developed as a sign of the gallic and catechol tannins.

Test for saponins: a test tube containing the crude extract and 5 mL of distilled water was rapidly shaken, resulting in the creation of a 1 cm foam, which denotes the presence of saponin.

The Salkowski test: To demonstrate the presence of terpenoids, 5 mL of the extract was mixed with 3 mL of concentrated H₂SO₄ and 2 mL of chloroform to create a reddish-brown layer.

Test for quinine: The 1 mL extract was mixed with diluted NaOH. The presence of quinines is indicated by blue-green or red colouring.

Testing for coumarin: 10% NaOH and chloroform were added to the extract and observing for a yellow colour indicated the presence of coumarin.

Test of Phenolic Compounds: In 5 mL of distilled water, 500 mg of the extract was dissolved, and a few droplets of a 5% ferric chloride solution were added. The presence of phenolic compounds was indicated by the presence of a dark green colour [14].

Experimental animals and study approval

Twenty-four male Wistar albino rats, aged 1 - 2 months and weighing an average of 100 - 150 gm, were obtained from the animal house of the College of Pharmacy, University of Baghdad. Before treatment, the animals had a two-week acclimatisation period in a typical laboratory setting. Water and a regular diet were freely available to them. They were kept under normal conditions of temperature ($25 \pm 5^\circ\text{C}$), 12-hour cycles of light and dark and humidity. Ethical protocol guidelines established by Baghdad University, College of Pharmacy, Research Ethics Committee that approved the study protocols and ethics.

Experimental design

The animals used in this study were divided equally into four groups, each with six rats; they were treated as follows:

Group I: (negative control) received an oral daily dose of 1 mL/kg/day of distilled water (DW) for 20 weeks. Group II: (positive control, HCC induction), in which rats intraperitoneally (I.P.) were injected with only 70 mg/kg diethyl nitrosamine (DEN) once *per week* for ten continuous weeks of the experimental duration of 20 weeks [15]. Group III: received 250 mg/kg [16] of Iraqi *H. tiliaceus* L. ethyl acetate leaf extract as chronic oral administration with food [17] for five days *per week* for 20 weeks, with a subsequent intraperitoneal dose of DEN (70 mg/kg) once *per week* for ten continuous weeks [15]. Group IV: received 500 mg/kg [16] of *H. tiliaceus* L. ethyl acetate leaf extract as chronic oral administration with food [17] for five days *per week* for 20 weeks, with a subsequent intraperitoneal dose of DEN (70 mg/kg) once *per week* for ten continuous weeks [15]. Rats were sacrificed ten weeks following the last DEN injection, after 24 hours of the end of the experimental period (20 weeks) [15].

We selected the control groups to determine the impact of plant extract as a protective measure against HCC. This was done by creating a positive control group (HCC model) and a negative control group (no treatment). By comparing the rats that received the plant extract with the positive control rats and the negative control rats, we aimed to demonstrate the potential effects of the plant.

Blood collection and tissue homogenate preparation

Following the euthanization of the rat, blood samples were collected in clean test tubes, allowed

to clot, and centrifuged at room temperature to obtain serum and liver tissue obtained for tissue homogenization. The liver was quickly removed, rinsed in ice-cold buffer phosphate saline pH 7.4, weighed before homogenization, and phosphate buffer saline (PBS) added to the tissue in a ratio of 1:9 (1 g of tissue and 9 mL of PBS). The tissue was then homogenized by a homogenizer after placing the tube in a beaker with ice. The homogenate was then spun using a cold centrifuge for 20 minutes at 10000 rpm, and the supernatant was used to calculate the levels of glutathione (GSH), superoxide dismutase (SOD) and malondialdehyde (MDA).

Determination of serum biochemical parameters

To determine the levels of the enzymes aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP) and total serum bilirubin (TSB), the serum was separated by centrifugation for 20 minutes at 3600 revolutions *per minute* (rpm).

Determination of serum and tissue homogenate parameters

Using the ELISA technique, the quantities of serum tumour necrosis factor (TNF- α), interleukin-6 (IL-6), interleukin-1 (IL-1 β), transforming growth factor beta (TGF- β), alpha-fetoprotein (AFP) and alpha-fetoprotein-L3 (AFP-L3) were determined as well as the tissue homogenate supernatant for the estimation of glutathione (GSH), superoxide dismutase (SOD) and malondialdehyde (MDA) levels. The test plate was ready, and the serum, standards, and blank were added before being incubated for an hour at 37°C . Following the incubation period, stop solutions, chromogenic substrate, enzyme conjugate and biotinylated-specific antibodies were added. Using a microplate reader, the optical density (OD) at 450 nm was calculated.

Histopathological evaluation

The freshly extracted liver tissues were preserved for 48 hours in 4% formaldehyde in 10 mM phosphate buffer (pH 7.4) for the histological investigations. For microscopic analysis, sections with a 5 M thickness were put on glass slides and stained with hematoxylin and eosin (H and E). Under a microscope, the sections were examined for any abnormalities.

Statistical analysis

Data were analysed using graph prism version 10. The numerical data were expressed using the mean $\pm$ standard deviation of the mean. The one-way ANOVA test was performed to compare each group to all other groups, and the Tukey multiple comparison test was utilized to determine the statistical significance of each group's comparison to the induction group. All results were deemed significant if the p-value for any of the study's findings was less than 0.05 ($p < 0.05$).

Results and Discussion

Phytochemical Screening

The extract shows the following components by using qualitative phytochemical analyses using established techniques [11] (Table I).

Table I
Phytochemical Screening Tests of *Hibiscus tiliaceus* leaves extract

Phytochemicals	<i>Hibiscus tiliaceus</i> leaves extract
flavonoids	+
tannins	+
saponins	+
terpenoids	+
coumarin	+
quinine	+
alkaloids	+
phenols	+

Effect of ethyl acetate leaf extract of *Hibiscus tiliaceus* on (hepatic tissue injury marker) serum ALT, AST, ALP and bilirubin

Table II displays that DEN administration caused a substantial increase in liver function tests (*i.e.*, AST, ALT, ALP and bilirubin) concentrations in rats as compared with controls ($p < 0.05$), indicating liver damage.

Treatments with the ethyl acetate extract of *Hibiscus tiliaceus* L. leaves resulted in dose-dependent amelioration of hepatic toxicity and a dose-dependent improvement in liver functions (Table II).
Effect of the ethyl acetate leaf extract of Hibiscus tiliaceus on AFP and AFP-L3

Group II (rats have HCC induction by DEN) showed a significant ($p < 0.05$) elevation in the levels of AFP

(5.259 ± 1.004) compared to the normal control group (0.7113 ± 0.1597) and a reduction in anti-oxidant systems (*i.e.*, SOD and GSH levels) in the HCC-induced group (30.13 ± 9.621 and 42.32 ± 12.35 , respectively) compared to the normal control group (106.2 ± 16.39 and 426.1 ± 7.41 , respectively). In a dose-dependent manner, the ethyl acetate extract of *Hibiscus tiliaceus* leaves decreased MDA concentration and increased SOD and GSH activities (Table III).

To examine the modulatory effect of treatment with ethyl acetate leaf extract of *Hibiscus tiliaceus* on the pro-inflammatory cytokines (TNF- α , IL-1 β and IL-6), levels were determined. DEN administration increases the levels of inflammatory cytokines (TNF-, IL-1 and IL-6; 2276 ± 364.9 , 7.363 ± 2.016 and 75.17 ± 7.069 , respectively; 367.0 ± 157.4 , 0.7980 ± 0.2494 and 9.937 ± 1.413). In contrast, treatment with ethyl acetate leaf extract of *Hibiscus tiliaceus* significantly restored the concentrations of inflammatory cytokines (TNF-, IL-1 and IL-6) levels, particularly at the higher dose (500 mg/kg) (1078 ± 518.3 , 1.097 ± 0.1208 and 8.322 ± 1.701 , respectively).

In addition, Figure 1 demonstrates an increase in TGF- β in the liver tissues of rats exposed to DEN (342.1 ± 82.67 , $p < 0.05$) compared to the normal control (59.57 ± 16.20 , $p < 0.05$). Similarly, administration of the ethyl acetate leaf extract of *Hibiscus tiliaceus* had the opposite effect by downregulating the TGF- β level dose-dependently, particularly at 500 mg/kg (111.3 ± 16.82 , $p < 0.05$).

Table II
Effect of Ethyl Acetate Iraqi *Hibiscus tiliaceus* leaves extract treatment on liver function tests in

Parameter	Control	DEN Induction	Ethyl acetate extract 250 mg/kg Treatment	Ethyl acetate extract 500 mg/kg Treatment
AST (U/L)	22.62 ± 1.397	$51.03 \pm 5.756^*$	$30.95 \pm 3.593^{\#}$	$19.38 \pm 0.7958^{\#}$
ALT (U/L)	27.32 ± 1.364	$60.16 \pm 8.001^*$	$37.06 \pm 5.459^{\#}$	$21.50 \pm 0.9296^{\#}$
ALP (U/L)	71.09 ± 4.508	$154.6 \pm 11.41^*$	$105.6 \pm 21.12^{\#}$	$90.70 \pm 4.250^{\#}$
Bilirubin (mg/dL)	0.3982 ± 0.02602	$1.590 \pm 0.2070^*$	$0.5660 \pm 0.06358^{\#}$	$0.3865 \pm 0.008408^{\#}$

and AFP-L3 compared with the corresponding levels in negative control group I rats.

The ethyl acetate leaf extract administration of *Hibiscus tiliaceus* decreased AFP and AFP-L3 levels compared to the DEN-induced HCC group (Figure 1).

Effect of ethyl acetate leaf extract of *Hibiscus tiliaceus* on enzymatic antioxidant levels and pro-inflammatory cytokines

Table III demonstrates a significant increase in oxidative stress markers, including MDA levels

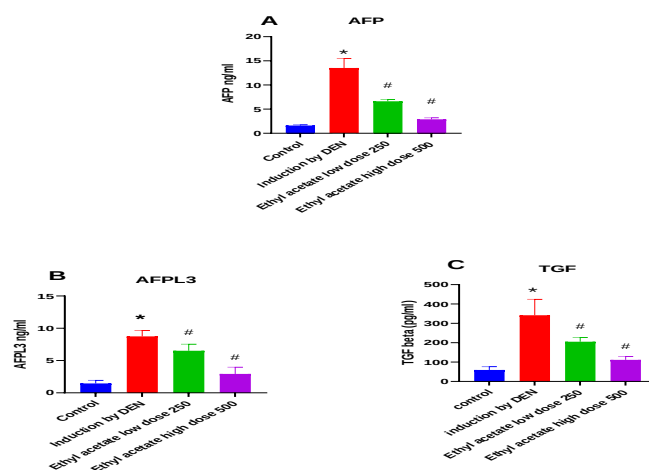
study groups

*denotes significant differences compared with the normal control group ($p < 0.05$);

[#]represents significant differences compared with the DEN group ($p < 0.05$).

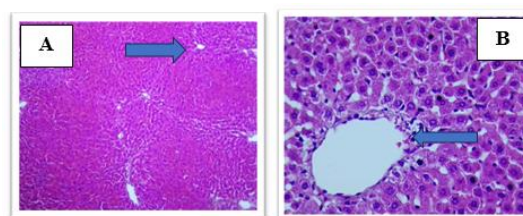
Table III
Effect of Ethyl Acetate Iraqi *Hibiscus tiliaceus* leaves extract treatment on enzymatic antioxidant levels and pro-inflammatory cytokines

Parameter	Control	DEN Induction	Ethyl acetate extract 250 mg/kg Treatment	Ethyl acetate extract 500 mg/kg Treatment
MDA (nmol/mL)	0.7113 ± 0.1597	$5.259 \pm 1.004^*$	$3.408 \pm 0.8575^{\#}$	$0.6365 \pm 0.06028^{\#}$
SOD (U/mL)	106.2 ± 16.39	$30.13 \pm 9.621^*$	$44.33 \pm 3.147^{\#}$	$94.10 \pm 3.634^{\#}$
GSH (ng/L)	426.1 ± 37.41	$42.32 \pm 12.35^*$	$280.0 \pm 85.16^{\#}$	$392.9 \pm 15.91^{\#}$
TNF- α (pg/mL)	367.0 ± 157.4	$2276 \pm 364.9^*$	1914 ± 324.4	$1078 \pm 518.3^{\#}$
IL-1 β (pg/mL)	0.7980 ± 0.2494	$7.363 \pm 2.016^*$	5.396 ± 1.466	$1.097 \pm 0.1208^{\#}$
IL-6 (pg/mL)	9.937 ± 1.413	$75.17 \pm 7.069^*$	$31.41 \pm 8.743^{\#}$	$8.322 \pm 1.701^{\#}$

**Figure 1.**

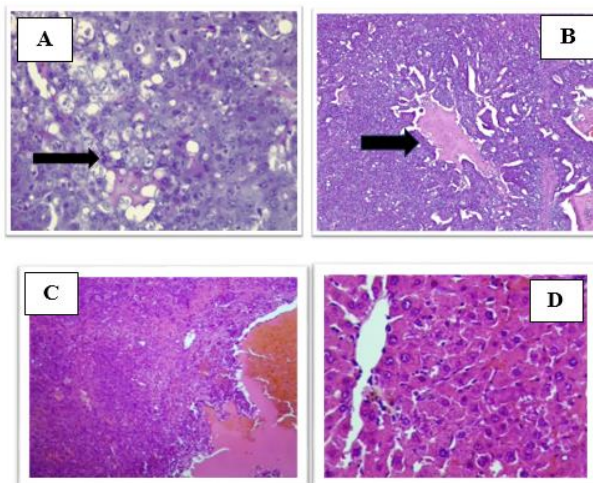
(A) AFP level, (B) AFP-L3 level and (C) TGF-β level

*Denotes significant differences compared with the normal control group ($p < 0.05$); #represents significant differences compared with the DEN group ($p < 0.05$).

**Figure 2.**

Liver tissue of negative control group rats

A and B) the normal architecture of the hepatic cord (the arrow shows the central vein) H and E (x100)

**Figure 3.**

Liver tissue of rats induced for HCC by DEN. H and E (x100)

A: The observed phenomenon in the image is characterised by the ballooning degeneration of multiple hepatocytes (black arrow). There is a presence of colloid formation in multiple areas, accompanied by a complete loss of the original liver architecture. B: There is a complete loss of the normal architecture of the liver. The central vein exhibits rupture and disruption, resulting in the absence of the typical arrangement of hepatic cords. Hepatocyte inflammation refers to the presence of fibrous tissue within the liver. C: The observed findings include a focal area of necrosis and inflammation of hepatocytes. D: There are indications of mitosis, the presence of polymorphic cells, and the presence of giant cells, which are characteristic of liver carcinoma.

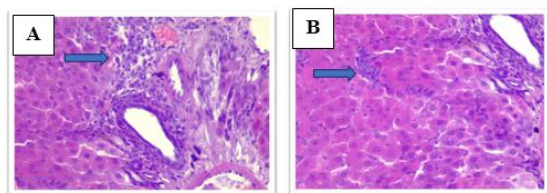


Figure 4.

Liver tissue of rats treated with 250 mg/kg *Hibiscus tiliaceus* ethyl acetate leaf extract H and E (x100)
A and B: liver sections demonstrate areas with hepatic necrosis replaced by mononuclear leukocytes (arrows)

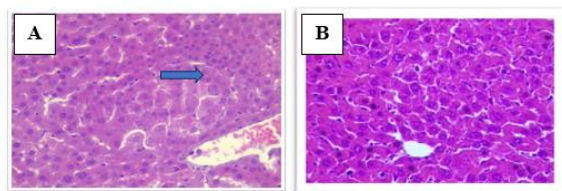


Figure 5.

Liver tissue of rats treated with 500 mg/kg *Hibiscus tiliaceus* ethyl acetate leaf extract H and E (x100)
A: Hepatocytes replaced all the degenerative cells not yet arranged in cords. Different sizes of cells and nuclei reflect different underlying cytological processes (arrow), some with diplo-nuclei.
B: Exhibiting the absence of ballooning, inflammatory cells and regeneration of hepatocytes around the central vein towards a near-normal liver architecture.

Liver Histopathological examination

The results of the H and E staining are shown in Figure 2. Normal liver histology, including the central vein, hepatic lobules and portal triads, was observed upon examination of liver sections from the control group (this group of rats was fed only distilled water and was regarded as a negative control). Both hepatic cords and hepatic sinusoids were arranged and preserved ordinarily, indicating that liver tissues from healthy controls exhibited no changes in histological architecture. DEN-treated rats showed a lack of typical hepatocyte architecture and lobules of neoplasm, cellular infiltration with areas of large fatty tumour cells characterised by irregular nuclei and massive changes in lipid accumulation, sinusoidal dilatation, necrosis and numerous mitotic figures, as depicted in Figure 3 of the DEN positive control group. Cirrhosis occurs when diffuse chronic hepatic injury results in the formation of fibrous septa that extend between the hepatic tracts, causing irreversible devastation of the normal hepatic architecture and the development of regenerative nodules in the liver parenchyma. With the development of portal hypertension, both the vascular inflow and outflow patterns were atypical and disrupted. The capsule's bumpy appearance was the liver's coarse external appearance. The average size of the regenerative hepatocyte nodules was less than 1 mm.

Histological examination of tissue sections from the livers of rats in group III that received 250 mg/kg of the ethyl acetate extract from the leaves of *Hibiscus tiliaceus* revealed limited necrotic and degenerative alterations (Figure 4).

Group IV received 500 mg/kg of *H. tiliaceus* L. ethyl acetate leaf extract, displaying less necrotic tissue than the group without treatment. The main hepatic architecture became more noticeable. The normal liver architecture was well preserved and observable, as shown in Figure 5.

The liver is a metabolically active organ that, in addition to biotransformation and clearance of endogenous compounds, also metabolises xenobiotics [18, 19]. Hepatocellular carcinoma, the most prevalent form of primary liver cancer, is characterised by a rising incidence and a high mortality rate. Animal models of hepatocellular carcinoma are frequently used to study the biology of cancer and evaluate new treatments. The rat model of DEN-induced hepatocellular carcinoma mimics the pathogenesis of human hepatocellular carcinoma, including liver damage, chronic inflammation, hepatocyte proliferation, liver fibrosis and cirrhosis, disorganised vasculature and modulations of the immune microenvironment. Preclinical drug testing using the DEN-induced rat liver cancer model [20]. Hepatocarcinogenesis is induced by the widely used hepatocarcinogen DEN, which influences cancer initiation by inducing DNA strand breaks and carcinogen adducts through alkylation [21]. DEN-induced hepatocellular carcinogenesis is also influenced by reactive oxygen species (ROS) [22].

H. tiliaceus possesses hepatoprotective and antimutagenic properties; it is a source of natural antimicrobial agents and antioxidants and has medical and therapeutic benefits; consequently, it contains an abundance of phytochemicals, including terpenoids, alkaloids, phenols and flavonoids [11, 23].

The liver function tests consist of alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP) and total bilirubin (TBS). These tests can detect liver injury, and the pattern of increased levels can aid in differential diagnosis [23]. When cells are damaged, the cytoplasmic enzymes aspartate transaminase (AST), alanine transaminase (ALT) and alkaline phosphatase (ALP) are released into the circulation. The release of these enzymes into the bloodstream in response to cell death and hepatocellular injury is regarded as a reliable indicator of the destruction of liver tissue and the regeneration of liver cells [24, 25]. They are commonly used in animal research to identify and monitor the development of hepatocarcinogenesis [26]. According to our findings in the DEN-induced HCC rat model, group II exhibited substantially elevated levels of ALT, AST, ALP and TSB

compared to the normal control group, which is consistent with other studies [27, 28]. The administration of *H. tiliaceus* leaf ethyl acetate extract decreased the levels of ALT, AST, ALP and TSB in a dose-dependent manner as compared to HCC-induced rats in group II, bringing them closer to those of normal control rats at a dose of 500 mg/kg.

AFP, a prominent glycoprotein with a molecular weight of 70 kDa, is a serum marker for predicting HCC and is a typical gold standard [29]. It is primarily generated by the foetal digestive tract, vital line sac cells and embryonic liver during the first trimester of pregnancy. After birth, the blood concentration of AFP rapidly decreases, and its expression is suppressed in adulthood. AFP is increased pathologically in HCC [30]. Our results indicate that the group II model of DEN-induced HCC rats had substantially higher AFP levels than the normal control group [27, 28]. AFP is essential for regulating tumor development and cell separation, as well as the proliferation of hepatoma cells, presumably via AFP receptors. In the DEN-induced group, elevated AFP levels demonstrated the presence of hepatocellular carcinoma [31].

In a dose-dependent manner, administration of *H. tiliaceus* leaf ethyl acetate extract caused a significant decrease in AFP level compared to HCC-induced rats in group II.

AFP can be elevated not only in HCC but also in pregnancy and chronic liver inflammation, which may result in false-positive or false-negative results. In recent years, AFP-L3 has been used as the specific tumor marker for HCC, though not as frequently as AFP alone [32]. The AFP-L3-positive HCC may exhibit accelerated growth and early metastasis [33]. Wistar rats exposed to the hepatocarcinogen DEN exhibited elevated levels of AFP-L3, whereas *H. tiliaceus* leaf ethyl acetate extract significantly decreased AFP-L3 levels relative to the DEN-initiated liver cancer group in a dose-dependent manner, with the higher dose becoming too close to the normal control rats.

TGF- β is a multifunctional cytokine that regulates the growth, migration, differentiation and apoptosis of various epithelial and hematopoietic cells. It is a member of the protein superfamily known as the TGF- β superfamily [34]. TGF- β elevation correlates directly with tissue injury severity and hepatic fibrosis [35]. By modifying the expression of TGF- β receptors, malignant cells can frequently reduce their TGF- β suppressive signaling. TGF- β regulates diverse inflammatory processes, inhibiting cellular activities such as proliferation, differentiation, and carcinogenesis survival [36]. Recently, it was asserted that TGF- β increased the potential for enhanced chemoresistance by inducing the expression of drug efflux transporters *via* activation of the nuclear PXR receptor for xenobiotics [37].

Group II in the DEN-induced HCC rat model had substantially higher levels of the cytokine TGF- β than the normal control group, consistent with previous research [38]. Compared to HCC-induced rats in group II, administration of *H. tiliaceus* leaf ethyl acetate extract significantly decreased the level of TGF- β in a dose-dependent manner.

Therefore, TGF- β and AFP-L3 may serve as prognostic markers for HCC, and they could be used to monitor the treatment response of HCC [34, 38].

The oxidative stress in the liver was increased by DEN-induced HCC, and the endogenous antioxidant enzymes were altered. DEN diminished the endogenous antioxidant defence system by significantly increasing the level of MDA (oxidative stress marker) and decreasing the levels of SOD and GSH (free radical scavenger antioxidant) [28].

Malondialdehyde levels are a common indicator of oxidative stress and antioxidant status in cancer patients [39, 40, 41]. Consistent with previous research, our study demonstrates that the DEN-induced HCC rat model in group II had substantially elevated MDA levels compared to the normal control group [15, 28]. Due to its high polyphenol, coumarin and flavonoid content, *H. tiliaceus* leaf ethyl acetate extract significantly decreased the level of MDA in a dose-dependent manner when compared to HCC-induced rats in group II [42, 43]. The most prominent antioxidant enzyme in aerobic cells, superoxide dismutase (SOD), eliminates superoxide radicals. SOD catalyses the decomposition of oxygen molecules and hydrogen peroxide. SOD can counteract the oxidant assault directly and protect cells from DNA damage [44].

Consistent with previous research, the DEN-induced HCC rat model in group II demonstrated a significant decrease in SOD and GSH levels compared to the normal control group [27, 28].

At a dose of 500 mg/kg, administration of *H. tiliaceus* leaf ethyl acetate extract significantly increased SOD and GSH levels due to the coumarins and flavonoid content of the extract [45, 46].

Since chronic inflammation increases cancer risk and influences every phase of tumorigenesis, from initiation to tumour promotion to metastasis, it has a profound effect on cancer development [47].

The production of proinflammatory TNF and IL1- β in response to tissue damage is linked to an acceleration of the cell cycle, oxidative stress-induced DNA damage and regulation of tumour development in HCC and the effect of IL-6 in the tumour microenvironment, which promotes cancer cell proliferation and contributes to HCC formation [48]. Our findings are similar to prior research in that group II in the DEN-induced HCC rat model displayed significantly higher levels of proinflammatory TNF- α , IL-6 and IL-1 β than the healthy control group [27, 28]. The administration of *H. tiliaceus* leaf ethyl

acetate extract caused a significant decrease in the level of IL-6 in a dose-dependent manner as compared with HCC-induced rats in group II that are too close to the normal control rats at a dose of 500 mg/kg and caused a significant decrease in the levels of TNF- α and IL-1 β when administered only at a dose of 500 mg/kg as compared with HCC-induced rats in group II.

In rats exposed to DEN, histological changes showed cellular boundary loss, ballooning degeneration, infiltration of inflammatory mediators and localised nodular hyperplasia in the liver. Results match earlier research [49].

In this study, we demonstrated that the injection of DEN into rats led to a marked elevation in the levels of serum AST, ALT and ALP, which is indicative of hepatocellular damage, as previously reported. On the other hand, treatment with the extracts of the three studied varieties, garlic, curcuma and silymarin, decreased liver damage and decreased the level of these enzymes [28, 50].

Also, DEN causes an increase in oxidative stress and MDA with a subsequent decrease in anti-oxidant GSH and SOD, and the treatment with the extracts of the three studied varieties, garlic, curcuma and silymarin, decreases MDA and increases GSH and SOD [28, 50].

This study has certain limitations that should be taken into consideration. One limitation is the extended duration of dosing, which necessitates a large quantity of plant extract. Additionally, administering the plant extract with food poses another limitation, as it requires separating the rats into individual cages to ensure they receive the precise doses. However, these limitations can be overcome through diligent and meticulous work.

Conclusions

Our study provides evidence that the administration of ethyl acetate extract from *H. tiliaceus* leaf has a beneficial effect on hepatocellular carcinomas by decreasing AFP and AFP-L3. The extracts also demonstrated excellent activity in managing oxidative stress enzymes (MDA, SOD and GSH), reducing liver inflammation and hepatic dysfunction via amelioration of liver biochemical enzymes (AST, ALT, ALP and total bilirubin), and improving the histology of liver tissue by decreasing necrosis and apoptosis. Therefore, this extract can serve as a therapeutic approach for HCC.

Conflict of interest

The authors declare no conflict of interest.

References

1. Aziz TA, Ahmed ZA, Juma'a KM, Abdulrazzaq MH, Hussain SA, Study of the protective effects of

- benfotiamine against CCl₄-induced hepatotoxicity in rats. *Iraqi J Pharm Sci.*, 2009; 18: 47-53.
2. Jemal A, Bray F, Center MM, Ferlay J, Ward E, Forman D, Global cancer statistics. *C J Clin.*, 2011; 61(2): 69-90.
3. Kareem VS, Zalzal MH, Safranl Effect against Cyclophosphamide-Induced Liver Injury. *Iraqi J Pharm Sci.*, 2021; 30(2): 208-213.
4. Mokdad AA, Singal AG, Yopp AC, Liver Cancer. *Jama*, 2015; 314(24): 2701.
5. Shirakami Y, Gottesman ME, Blaner WS, Diethylnitrosamine-induced hepatocarcinogenesis is suppressed in lecithin: Retinol acyltransferase-deficient mice primarily through retinoid actions immediately after carcinogen administration. *Carcinogenesis*, 2012; 33(2): 268-274.
6. Park DH, Shin JW, Park SK, Seo JN, Li L, Jang JJ, Lee MJ, Diethylnitrosamine (DEN) induces irreversible hepatocellular carcinogenesis through overexpression of G1/S-phase regulatory proteins in rat. *Toxicol Lett.*, 2009; 191(2-3): 321-326.
7. Singh D, Singh M, Yadav E, Falls N, Komal U, Dangi DS, Kumar V, Verma A, Amelioration of diethylnitrosamine (DEN)-induced hepatocellular carcinogenesis in animal models via knockdown oxidative stress and proinflammatory markers by *Madhuca longifolia* embedded silver nanoparticles. *RSC Adv.*, 2018; 8(13): 6940-6953.
8. Al-Mashhadani Z, Naser RA, Zalzal MH, The impact of new targeting methods in the cancer therapy. *Int J Appl Pharm.*, 2019; 11(2): 1-6.
9. Iancu C, Filip N, Mocanu M, Dehelean CA, Danciu C, Cioancă O, Hăncianu M, Corciovă A, Burlec F, Mircea C, Robu S, Antioxidant and anticancer effect of some *Pelargonium* species extracts. *Farmacia*, 2023; 71(2): 312-321.
10. Vanzella C, Bianchetti P, Sbaraini S, Vanzin SI, Melecchi MI, Caramão EB, Siqueira IR, Antidepressant-like effects of methanol extract of *Hibiscus tiliaceus* flowers in mice. *BMC Compl Alter Med.*, 2012; 12: 1-6.
11. Rawool PP, Parulekar BC, Phytochemical screening of *Hibiscus tiliaceus* by FTIR spectroscopic analysis. *Inter J Phar Biol Sci.*, 2019; 9(3): 1308-1319.
12. Mahire SP, Patel SN, Extraction of phytochemicals and study of its antimicrobial and antioxidant activity of *Helicteres isora* L.. *Clin Phytosci.*, 2020; 6(1): 1-6.
13. Mutlag SH, Hamad MN, Abbas IS, Ismael SH, The evaluation of ethyl acetate fraction of *Cressa cretica* effect on mitotic index and micronucleous Frequency in Mice. *Int J Pharm Sci Rev Res.*, 2017; 45(28): 147-150.
14. Bankole AE, Adekunle AA, Sowemimo AA, Umebese CE, Abiodun O, Gbotosho GO, Phytochemical screening and *in vivo* antimalarial activity of extracts from three medicinal plants used in malaria treatment in Nigeria. *Parasitol Res.*, 2016; 115(1): 299-305.
15. Song J, Ding W, Liu B, Liu D, Xia Z, Zhang L, Cui L, Luo Y, Jia X, Feng L, Anticancer effect of caudatin in diethylnitrosamine-induced hepatocarcinogenesis in rats. *Mol Med Rep.*, 2020; 22(2): 697-706.

16. Narender KS, Kumar D, Kumar V, Antinociceptive and anti-inflammatory activity of *Hibiscus tiliaceus* leaves. *Inter J Pharmacog Phytochem Res.*, 2009; 1(1): 15-17.
17. Turrini F, Vallarino G, Cisani F, Donno D, Beccaro GL, Zunin P, Boggia R, Pittaluga A, Grilli M, Use of an Animal Model to Evaluate Anxiolytic Effects of Dietary Supplementation with *Tilia tomentosa* Moench Bud Extracts. *Nutrients*, 2020; 12(11): 3328.
18. Gao B, Jeong W Il, Tian Z, Liver: An organ with predominant innate immunity. *Hepatology*, 2008; 47(2): 729-736.
19. Ahmad HA, Kathem SH, Thymoquinone Attenuates Immune Mediated Liver Injury Induced by Concanavalin A in Mice. *Iraqi J Pharm Sci.*, 2021; 30(2): 50-57.
20. Matter MS, Marquardt JU, Andersen JB, Quintavalle C, Korokhov N, Stauffer JK, Kaji K, Decaens T, Quagliata L, Elloumi F, Hoang T, Molinolo A, Conner EA, Weber A, Heikenwalder M, Factor VM, Thorgeirsson SS, Oncogenic driver genes and the inflammatory microenvironment dictate liver tumor phenotype. *Hepatology*, 2016; 63(6): 1888-1899.
21. Tolba R, Kraus T, Liedtke C, Schwarz M, Weiskirchen R, Diethylnitrosamine (DEN)-induced carcinogenic liver injury in mice. *Lab Anim.*, 2015; 49:59-69.
22. Lin H, Liu X bo, Yu J jiao, Hua F, Hu Z wei, Antioxidant N-Acetylcysteine Attenuates Hepatocarcinogenesis by Inhibiting ROS/ER Stress in TLR2 Deficient Mouse. *PLoS One.*, 2013; 8(10).
23. Rajeswari G, Priyanka B, Amrutha RE, Rajaram C, Kanhere RS, Kumar SN, *Hibiscus tiliaceus*: a possible immunomodulatory agent. *J Pharm Res.*, 2013; 6(7): 742-747.
24. Ribeiro AJS, Yang X, Patel V, Madabushi R, Strauss DG, Liver Microphysiological Systems for Predicting and Evaluating Drug Effects. *Clin Pharmacol Ther.*, 2019; 106(1): 139-147.
25. Prati D, Taioli E, Zanella A, Della Torre E, Butelli S, Del Vecchio E, Vianello L, Zanuso F, Mozzi F, Milani S, Conte D, Colombo M, Sirchia G, Updated definitions of healthy ranges for serum alanine aminotransferase levels. *Ann Intern Med.*, 2002; 137(1): 1-10.
26. Oh RC, Hustead TR, Ali SM, Pantsari MW, Mildly elevated liver transaminase levels: causes and evaluation. *Amer Fam Phys.*, 2017; 96(11): 709-715.
27. You Y, Zhu F, Li Z, Zhang L, Xie Y, Chinnathambi A, Alahmadi TA, Lu B, Phyllanthin prevents diethylnitrosamine (DEN) induced liver carcinogenesis in rats and induces apoptotic cell death in HepG2 cells. *Biomed Pharmacother.*, 2021; 137: 111335.
28. Zhang Y, Li X, Li X, Curcuma ameliorates diethylnitrosamine-induced hepatocellular carcinoma via alteration of oxidative stress, inflammation and gut microbiota. *J Inflamm Res.*, 2021; 14: 5551.
29. Gillespie JR, Uversky VN, Structure and function of α -fetoprotein: a biophysical overview. *Biochim Biophys Acta - Protein Struct Mol Enzymol.*, 2000; 1480(1-2): 41-56.
30. Omar MA, Omran MM, Farid K, Tabll AA, Shahein YE, Emran TM, Petrovic A, Lucic NR, Smolic R, Kovac T, Smolic M, Biomarkers for Hepatocellular Carcinoma: From Origin to Clinical Diagnosis. *Biomedicines*, 2023; 11(7): 1852.
31. Galle PR, Foerster F, Kudo M, Chan SL, Llovet JM, Qin S, Schelman WR, Chintharlapalli S, Abada PB, Sherman M, Zhu AX, Biology and significance of alpha-fetoprotein in hepatocellular carcinoma. *Liver Int.*, 2019; 39(12): 2214-2229.
32. Li D, Mallory T, Satomura S, AFP-L3: a new generation of tumor marker for hepatocellular carcinoma. *Clin Chim Acta.*, 2001; 313(1-2): 15-19.
33. Force M, Park G, Chalikonda D, Roth C, Cohen M, Haleboua-DeMarzio D, Hann HW, Alpha-Fetoprotein (AFP) and AFP-L3 Is Most Useful in Detection of Recurrence of Hepatocellular Carcinoma in Patients after Tumor Ablation and with Low AFP Level. *Viruses*, 2022; 14(4): 775.
34. Chen J, Gingold JA, Su X, Immunomodulatory TGF- β signaling in hepatocellular carcinoma. *Trends Mol Med.*, 2019; 25(11): 1010-1023.
35. Shawky Elsawabi A, Abdel wahab K, Ibrahim W, Saleh S, Massoud Y, Abdelbary M, Nabih A, α -Fetoprotein (AFP)-L3% and transforming growth factor B1 (TGFB1) in prognosis of hepatocellular carcinoma after radiofrequency. *Egypt Liver J.*, 2019; 9(1).
36. Massagué J, TGF β in cancer. *Cell*, 2008; 134(2): 215-230.
37. Bhagyaraj E, Ahuja N, Kumar S, Tiwari D, Gupta S, Nanduri R, Gupta P, TGF- β induced chemoresistance in liver cancer is modulated by xenobiotic nuclear receptor PXR. *Cell Cycle*, 2019; 18(24): 3589-3602.
38. Shen Y, Wei Y, Wang Z, Jing Y, He H, Yuan J, Li R, Zhao Q, Wei L, Yang T, Lu J, TGF- β regulates hepatocellular carcinoma progression by inducing Treg cell polarization. *Cell Physiol Biochem.*, 2015; 35(4): 1623-1632.
39. Fu Y, Chung F-L, Oxidative stress and hepatocarcinogenesis. *Hepatoma Res.*, 2018; 4(8): 39.
40. Alduboni NM, Al-Shawi NN, Possible protective effects of two different doses of cyanocobalamin against methotrexate nephrotoxicity in rats. *Iraqi J Pham Sci.*, 2022; 31(2): 212-217.
41. Abdul-Wahab FK, Al-Shawi NN, Effects of vitamin D3 on methotrexate-induced jejunum damage in rats. *J Pharm Sci.*, 2020; 29(1): 260-267.
42. Wong SK, Lim YY, Chan EW, Antioxidant properties of *Hibiscus*: Species variation, altitudinal change, coastal influence and floral colour change. *J Trop For Sci.*, 2009; 21(4): 307-315.
43. Samsudin MS, Andriani Y, Sarjono PR, Syamsumir DF, Study on *Hibiscus tiliaceus* leaves as antibacterial and antioxidant agents. *Alotrop.*, 2019; 3(2): 123-131.
44. Gurudath S, Naik RM, Ganapathy KS, Guruprasad Y, Sujatha D, Pai A, Superoxide dismutase and glutathione peroxidase in oral submucous fibrosis, oral leukoplakia, and oral cancer: A comparative study. *J Orofac Sci.*, 2012; 4(2): 114-119.
45. Rosa RM, Moura DJ, Melecchi MI, dos Santos RS, Richter MF, Camarao EB, Henriques JA, de Paula Ramos AL, Saffi J, Protective effects of *Hibiscus tiliaceus* L. methanolic extract to V79 cells against

- cytotoxicity and genotoxicity induced by hydrogen peroxide and tert-butyl-hydroperoxide. *Toxicol In Vitro.*, 2007; 21(8): 1442-1452.
46. Muttlag SH, Kasim AA, Jassim AR, The Possible Protective Effect of Simultaneous Administration of Alcoholic *Ammi majus* Seeds' Extract Against Gentamicin induced Nephrotoxicity in Rats. *Al Mustansiriyah J Pham Sci.*, 2012; 11(1): 71-78.
 47. Grivennikov SI, Greten FR, Karin M, Immunity, inflammation, and cancer. *Cell*, 2010; 140(6) :883-899.
 48. Rico Montanari N, Anugwom CM, Boonstra A, Debes JD, The role of cytokines in the different stages of hepatocellular carcinoma. *Cancers*, 2021; 13(19): 4876.
 49. Kurma K, Manches O, Chuffart F, Sturm N, Gharzeddine K, Zhang J, Mercey-Ressejac M, Rousseaux S, Millet A, Lerat H, Marche PN, DEN-induced rat model reproduces key features of human hepatocellular carcinoma. *Cancers*, 2021; 13(19): 4981.
 50. Shaarawy SM, Tohamy AA, Elgendy SM, Abd Elmageed ZY, Bahnasy A, Mohamed MS, Kandil E, Matrougui K, Protective effects of garlic and silymarin on NDEA-induced rats hepatotoxicity. *Inter J Biol Sci.*, 2009; 5(6): 549.